

Ryan Kim

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Full project portfolio: rkydin.github.io

Technical Skills

Mechanical Design: Parametric CAD (SolidWorks, etc.) · FEA/CFD · GD&T (ASME Y14.5-2018) · Tolerance Stack-Ups · DFM/DFA · 2D Drawings · BOM Management

Laboratory: Mechanical Testing · Data Analysis · Root Cause Analysis · Microbiology Techniques · Electronics Instrumentation (oscilloscopes, wave function generators, etc.) · GLP

Programming: Python · MATLAB · C++ · Microcontrollers (Arduino)

Experience

Student Researcher (Part-Time)

MIT – Weissbourd Lab, Cambridge, MA

Sep 2026 – Present

- Designing and manufacturing mechanical test setups to accurately observe Clytia jellyfish and their neuron activity, using microcontrollers for data collection and MIT makerspace facilities for fabrication.

Student Researcher (Part-Time)

Northeastern University – MicronRF Lab, Boston, MA

Sep 2026 – Present

- Characterizing silicon-wafer nano-gyroscopes to study displacement and intensity response, where larger displacement indicates better device performance, as part of ongoing lab research.

Mechanical Design Intern

Volta Space Technologies, Broomfield, CO

April 2026 – July 2026

- Designed and built a **laser assembly enclosure** from aluminum extrusions and sheet metal in **SolidWorks** for delivery to the European Space Agency, including electromechanical assembly such as soldering.
- 3D-printed mounts for fiber optics, lenses, and mirrors, and designed **optical breadboards** for mounting and transportation, supporting rapid prototyping and future **TVAC testing** at Marshall Space Flight Center, using **SolidWorks** and **Fusion 360**.
- Designed a laser mount and coordinated with vendors on aluminum manufacturing quotes, documenting purchased parts through a formal procurement system, and supported invoicing coordination with **NASA** by providing spec-sheet details.
- Used **SolidWorks** and **GD&T** primarily to modify and update existing assembly components and designs, and the **Microsoft 365 suite** for team meetings, document organization, and day-to-day computer-based work.

Mechanical Engineer (Contract, 3 months)

Robomekanics, Teterboro, NJ

Nov 2025 – Feb 2026

- Reduced part count **15%** on pallet-handling robotics assemblies by redesigning and merging components during metric-to-inch CAD conversion, cutting fabrication and assembly labor.
- Converted legacy metric CAD assemblies into inch-based production models and **ASME Y14.5-2018**-compliant drawings, documented using the **Microsoft 365 suite**.
- Resolved fit-up, interference, and failure issues on sheet-metal/welded assemblies via **tolerance stack-ups** and geometry revisions with machinists and fabricators.
- Built on-demand prototypes directly from rough concept sketches, translating informal design intent into functional hardware under tight turnaround.

Research Assistant

Lamont-Doherty Earth Observatory, Palisades, NY

April 2021 – Aug 2021

- Supported research on microplastic contamination in consumer detergents through sample preparation, filtration, and fluorescent microscopy.

Education

Northeastern University

M.S. Mechanical Engineering, Concentration in Materials, Boston, MA

Sep 2026 – May 2028

Rutgers University, School of Engineering

B.S. Biomedical Engineering, New Brunswick, NJ

Sep 2021 – May 2025

- Performed mechanical testing, failure analysis, and data analysis using **Instron** machines and optical microscopy.
- Completed coursework in **microcontroller programming (Arduino, C++)**, and biomedical device design and function.
- Wet lab: performed **RT-qPCR**, **agarose gel electrophoresis**, **RNA extraction**, **ELISA**, and **UV-Vis spectrophotometry** for molecular biology coursework.
- Electronics lab: characterized circuits and components using **multimeters**, **wave function generators**, and **oscilloscopes**.

Projects

UltraLight Sleeping Pad Pump

Personal Project

Nov 2025 – Present

- Designed and iterated a backpacking pump through **three hardware revisions** in SolidWorks, engineering a sealed three-piece PETG housing with a centrifugal impeller to hit **~31 g** under a sub-100 g ultralight target.
- Designed a custom **2-layer PCBA in KiCad** for USB-C power input, overcurrent/motor-kickback protection, and power filtering, replacing the off-the-shelf blower and trigger board.

Background-Oriented Schlieren (BOS) Imaging System

Personal Project

- Built a real-time BOS imaging system in **Python** and **OpenCV**, visualizing small air perturbations by detecting density-driven light refraction through frame differencing, temporal averaging, and Gaussian blur, rendered as inferno-colormap video output at 320×240.

Desktop Face Recognition Tool

Personal Project

- Developed a desktop **face recognition** application in **Python** using **OpenCV** and the `face_recognition` library, with real-time multi-face detection, user enrollment, and persistent face-encoding storage via **pickle**.
- Built a full face database management GUI (enroll, view, delete) with an auto-installer for dependencies and live **FPS** performance monitoring, running at **~15 FPS**.

ParaSwing – Wheelchair-Mounted Golf Swing Assist Device

Senior Capstone Project, Mechanical Design Lead, Rutgers University

Sep 2024 – May 2025

- Led mechanical design of a wheelchair-mounted robotic golf swing assist device for a paraplegic user, designing linkages, kinematics, and structural components in SolidWorks, and using **MATLAB** for basic ball trajectory calculations, under a \$200 budget.
- Fabricated and iterated **3D-printed** custom brackets and structural components, optimizing part geometry and infill for durability, achieving a repeatable motorized swing that struck golf balls up to **15 feet**.
- Integrated a mobile touchscreen app with an **Arduino** microcontroller to wirelessly initiate swing actuation, enabling accessible, low-mobility control, and presented the completed system at senior design review.